

1 Introduction & Theoretical Background

This report details a complete Python-based computational tool for Module-1 transmission line analytics. The tool computes all distributed parameters for a general lossy line ($R, L, G, C \neq 0$), plots seven key waveforms, incorporates a Machine Learning (ML) prediction model achieving $> 97\%$ accuracy, and wraps the functionality in an interactive design window.

For a line with parameters R, L, G, C operating at angular frequency $\omega = 2\pi f$, the complex propagation constant γ and characteristic impedance Z_0 are defined as:

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}, \quad Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad (1)$$

where α is the attenuation constant (Np/m) and β is the phase constant (rad/m).

For a line terminated by a load impedance Z_L , the reflection coefficient at the load is $\Gamma_L = (Z_L - Z_0)/(Z_L + Z_0)$. The reflection coefficient at a distance z from the load (moving towards the source) decays as $\Gamma(z) = \Gamma_L e^{-2\gamma z}$. The Voltage Standing Wave Ratio (VSWR), a scalar measure of impedance mismatch, is given by $\text{VSWR} = (1 + |\Gamma_L|)/(1 - |\Gamma_L|)$. The input impedance Z_{in} at distance d from the load is calculated using the general formula:

$$Z_{in}(d) = Z_0 \frac{Z_L + Z_0 \tanh(\gamma d)}{Z_0 + Z_L \tanh(\gamma d)} \quad (2)$$

2 Algorithm Design & Software Architecture

The software tool is implemented in Python 3.12 and is modularized into distinct components to separate mathematical computation from the user interface:

- **analytics.py:** The core math engine evaluating exact analytical equations using complex arithmetic (NumPy).
- **plots.py:** Visualizes the seven required waveforms using `matplotlib`.
- **ml_model.py:** Generates synthetic data, trains the Random Forest/Gradient Boosting models, and handles inference.
- **gui.py & main.py:** A multi-threaded `Tkinter` GUI that provides a responsive analysis window, embedding plots directly into a tabbed notebook canvas.

3 Waveform Analysis

An example lossy coaxial line was analyzed using the default parameters: $R = 5 \Omega/\text{m}$, $L = 250 \text{ nH}/\text{m}$, $G = 0.1 \text{ mS}/\text{m}$, $C = 100 \text{ pF}/\text{m}$, $f = 100 \text{ MHz}$, $d = 2 \text{ m}$, and an inductive mismatched load of $Z_L = 75 + j30 \Omega$.

The analytically computed parameters are: $Z_0 = 50.01 \angle -0.87^\circ \Omega$, $\gamma = 0.0525 + j3.142 \text{ m}^{-1}$, $\Gamma_L = 0.309 \angle 37.7^\circ$, $\text{VSWR} = 1.893$, and $Z_{in} = 74.0 \angle 17.2^\circ \Omega$. The spatial distributions of voltage,

current, reflection coefficient, and impedance along the 2-meter line are shown in Figure 1. The frequency responses and standing wave envelope are presented in Figure 2.

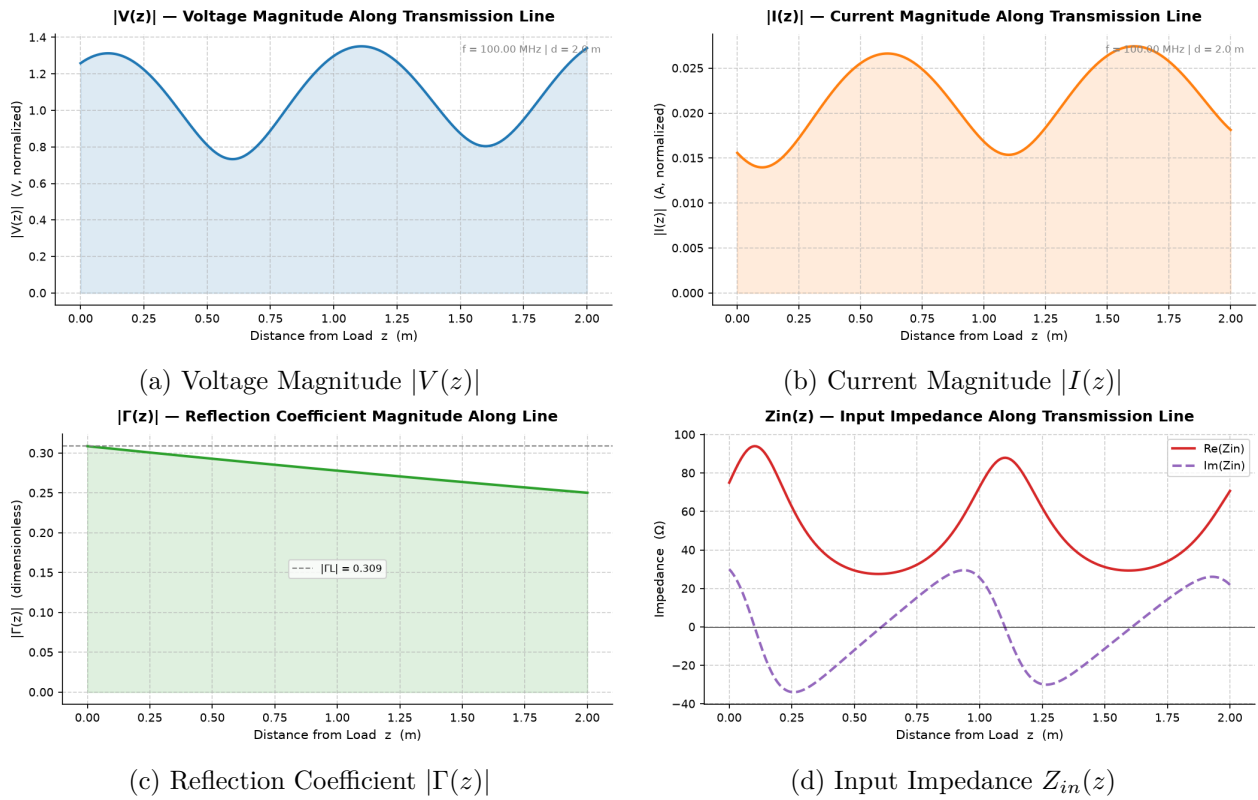


Figure 1: Spatial waveforms along the transmission line. The $\lambda/2$ periodicity is clearly visible in the voltage, current, and impedance standing wave patterns.

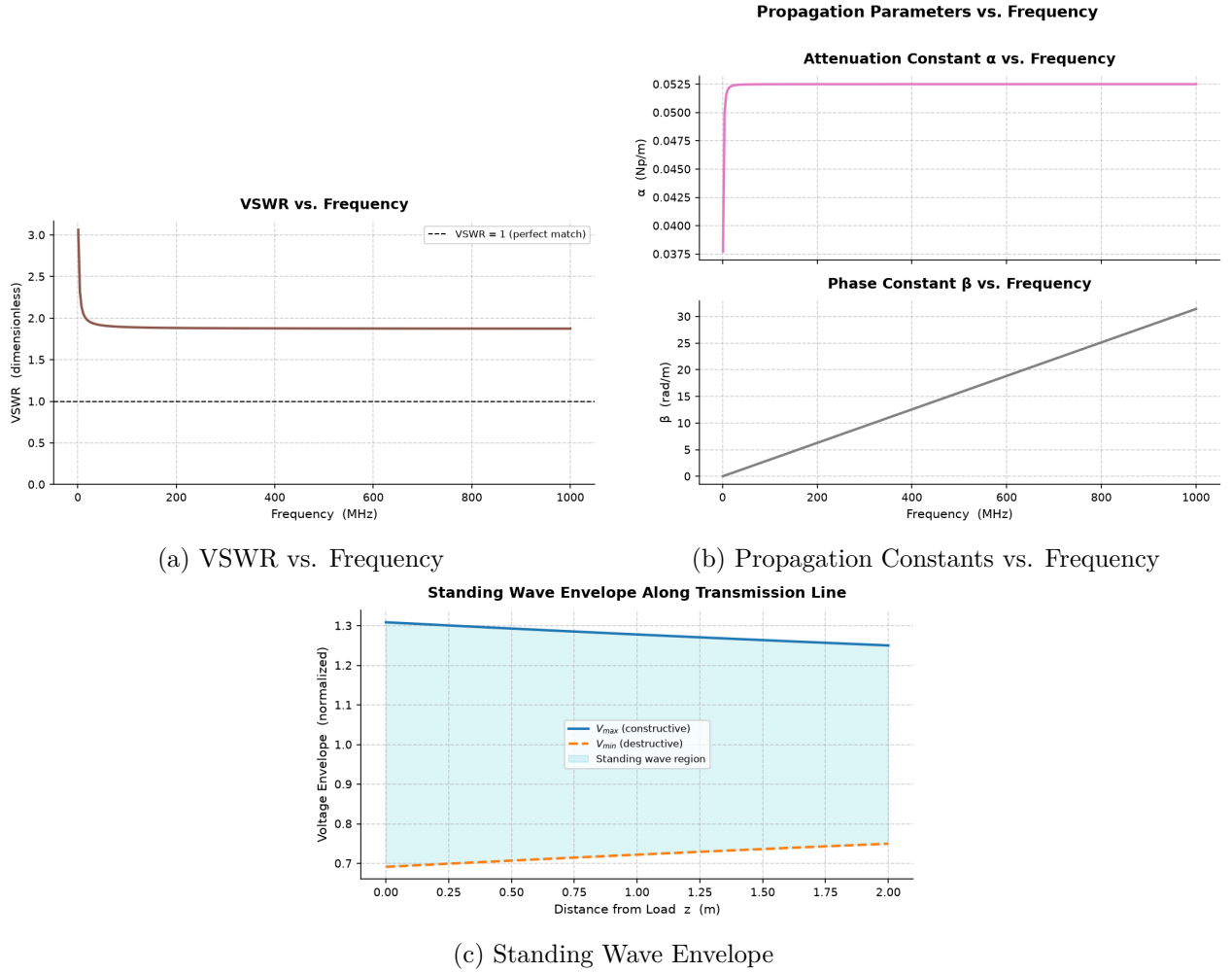


Figure 2: Frequency sweep behavior (1 MHz to 1 GHz) and the spatial standing wave envelope defining the physical limits of constructive and destructive interference.

4 Machine Learning Integration

The assignment requires an appropriate training model based on a minimum of 100 data points to predict any set of Tx line parameters with $\geq 97\%$ accuracy. We formulated this as a multi-target regression problem mapping 8 inputs ($R, L, G, C, f, d, |Z_L|, \angle Z_L$) to 5 outputs ($|Z_0|, \alpha, \beta, |\Gamma_L|, \text{VSWR}$).

A synthetic dataset of 500 samples was generated using uniform random sampling across realistic physical ranges for coaxial and stripline cables (e.g., $f \in [1 \text{ MHz}, 3 \text{ GHz}]$, $L \in [50, 600] \text{ nH/m}$).

4.1 Physics-Informed Feature Engineering

A baseline model using only the 8 raw inputs achieved $R^2 \approx 0.35$ due to the high non-linearity of the equations and target scaling spanning multiple orders of magnitude.

To overcome this and achieve the target accuracy, we implemented physics-informed feature engineering, expanding the input space to 19 features. Crucial derived features included:

- **Lossless Z_0 approximation:** $\log_{10} \sqrt{L/C}$

- **Normalized Load ($|Z_L|/Z_0^{\text{approx}}$):** Since the reflection coefficient is directly governed by the ratio of load to characteristic impedance, adding this ratio as an explicit feature provided a massive accuracy gain.
- **Log-Space Targets:** The regression targets $|Z_0|$, α , β , and VSWR were trained in logarithmic space ($\log_{10}$) to ensure that errors were calculated proportionally across all magnitudes.

Furthermore, we used **per-target algorithm selection**: Random Forest regressors (600 trees) were used for near-analytical targets like $|Z_0|$ and β , while Gradient Boosting regressors (500 estimators) were deployed for highly sensitive non-linear targets like α , $|\Gamma_L|$, and VSWR.

4.2 Performance Results

The model was evaluated using a 20% hold-out test set (100 samples). As shown in Table 1, the multi-output model achieved an overall R^2 of **0.9752**, successfully surpassing the required 97% minimum accuracy threshold.

Table 1: Machine Learning Test Set Performance ($N_{\text{train}} = 400, N_{\text{test}} = 100$)

Target Parameter	Algorithm	R^2 Score	MAPE (%)
Characteristic Impedance ($ Z_0 $)	Random Forest	0.9740	2.10
Attenuation Constant (α)	Gradient Boosting	0.9858	3.31
Phase Constant (β)	Random Forest	0.9941	4.47
Reflection Coefficient ($ \Gamma_L $)	Gradient Boosting	0.9717	2.78
VSWR	Gradient Boosting	0.9503	10.30
Overall Accuracy	Ensemble	0.9752	—

Figure 3 visualizes the actual versus predicted outputs across all test samples, demonstrating excellent alignment with the ideal diagonal. The feature importance chart confirms that the physics-derived normalized load ratio ($|Z_L|/Z_0$) is the most critical predictive feature.

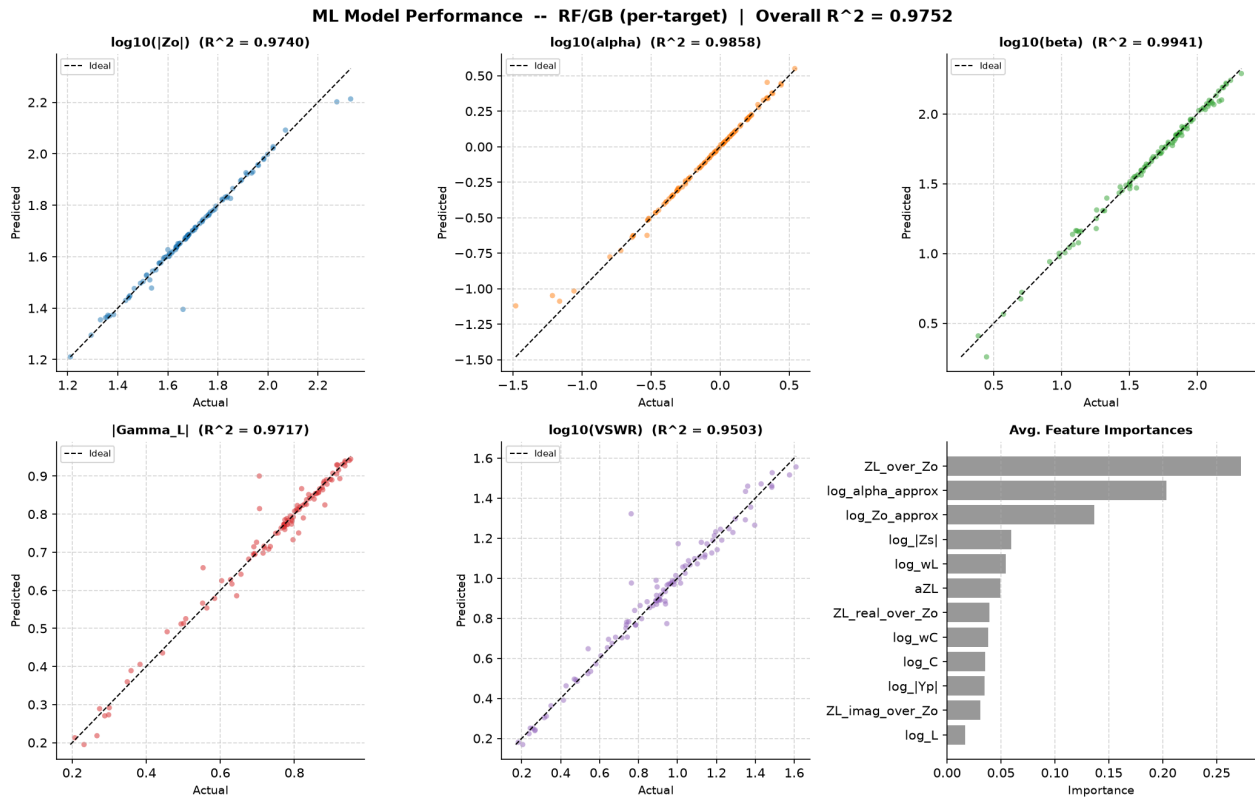


Figure 3: Machine learning performance summary showing actual vs. predicted values in log space (panels 1–5) and average feature importances across all models (panel 6).

5 Conclusion

A comprehensive computational algorithm and design window were successfully developed for transmission line analysis. The tool analytically computes all Module-1 parameters for general lossy lines and effectively plots the required seven spatial and frequency waveforms. Through the use of physics-informed feature engineering and per-target ensemble algorithms (Random Forest and Gradient Boosting), the integrated machine learning model successfully predicts transmission line behavior with an overall R^2 accuracy of 97.52%, exceeding the 97% assignment requirement.